

Olympic Moment: Decode Medal Success

Summary

The Paris 2024 Olympics delivered thrilling medal battles: the U.S. topped the charts with 126 medals, while China, together with U.S., matched their 40 golds. France and Great Britain flexed their depth, and underdogs like Dominica and St. Lucia made history with their first-ever golds. Yet, over 60 nations still await their Olympic moment. To decode medal success and identify key factors, we built a predictive model using historical Olympics data, exploring the celebrity “effect” of elite coaches and offering strategies to boost future performance.

For Task 1, we first **selected 8 indicators** based on **Pearson correlation coefficient**. Subsequently, we constructed a **Transformer-based Prediction Model** to predict the medal outcomes for each event. This model fully utilized the advantage of **attention mechanism** and provided a more fine-grained event-outcome estimation. Our model achieved a mean square error (MSE) of only **0.558**, indicating excellent performance. Using this model, we predicted the medal table in 2028 with prediction intervals, identified the progressing countries, and forecast 5 countries which are likely to win their first medal.

For Task 2, we employed a **Trend Decomposition Model Based on Locally Weighted Scatterplot Smoothing (LOESS)** to fit the trend of medal count for countries coached by Lang Ping and Béla Károlyi. Through this process, we found significant influence of the “great coach” effect and quantified the influence of a great coach. Subsequently, we constructed an **Evaluation of Future Prospects Model** to screen for events with investment potential from three dimensions, and selected three countries to assess the impact of introducing an outstanding coach. We found that recruitment of a world-class coach significantly affected a team’s ability to win medals.

For Task 3, we analyzed **in-depth and comprehensively** summarize of the Olympic medal scores based on the results from Task 1 and Task 2, and put forward targeted suggestions to country Olympic committees. Specifically, we suggested that host countries could increase their medal count by investing more on their advantageous events; small- and medium-sized countries should concentrate their resources on a few key events to achieve efficient resource utilization; and all countries should actively introduce excellent coaches to significantly enhance their teams’ competitive level and medal-winning potential.

Keywords: Transformer; Pearson correlation coefficient; LOESS;

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1 Introduction

1.1 Problem Background

“In the Olympics, anything can happen. The medal table is always full of surprises and uncertainties.”

The Olympic Games stand as the world’s **largest and most prestigious** multi-sport event, serving as a vital platform for nations to showcase their athletic excellence. The medal table ranking, in particular, is a key indicator of a country’s sporting prowess, **drawing the attention of athletes and spectators worldwide**.

At the 2024 Summer Olympics in Paris, the United States demonstrated its dominant strength by leading the overall medal count, while fiercely competing with China for the top position in the gold medal standings. Notably, first-time medal-winning nations such as Albania and Cabo Verde also captured significant global interest. However, with over 60 countries still awaiting their first Olympic medal, which underscores the persistent disparities in global sports development.

Against this backdrop, identifying the factors that influence a country’s Olympic medal performance and constructing an accurate predictive model for future medal counts becomes essential [1]. Such efforts are critically important for nations to achieve more effective preparation strategies and provides valuable insights into the broader societal dynamics that shape global sports development.

1.2 Restatement of the Problem

Given the background information, problem constraints, and additional guidance, we aim to accomplish the following tasks according to the provided data sets:

- Develop a predictive model to forecast each country’s future Olympic medal counts, incorporating uncertainty or precision estimates, and establish performance metrics for model evaluation. Using the developed model, fulfill the following objectives:
 - (i) Predict the medal table for the 2028 Olympics in Los Angeles, USA, including confidence intervals. Identify which countries are expected to improve and which are likely to decline.
 - (ii) Estimate the number of countries likely to win their first medal at the next Games, along with associated probabilities.
 - (iii) Examine the relationship between Olympic events and the medal outcomes of participating countries. Identify the most impactful sports for different nations and evaluate how the host country’s event selections influence overall medal performance.
- Create a model to evaluate the contribution of “great coaches” to medal counts. Select three countries, identify the sports where investing in a “great coach” would be most strategic, and estimate the potential impact.
- Derive original insights from our models regarding Olympic medal counts and explain how these insights can inform national Olympic committees.

1.3 Our Work

Figure 1 clearly shows the key steps and modeling approach to accomplish the tasks.

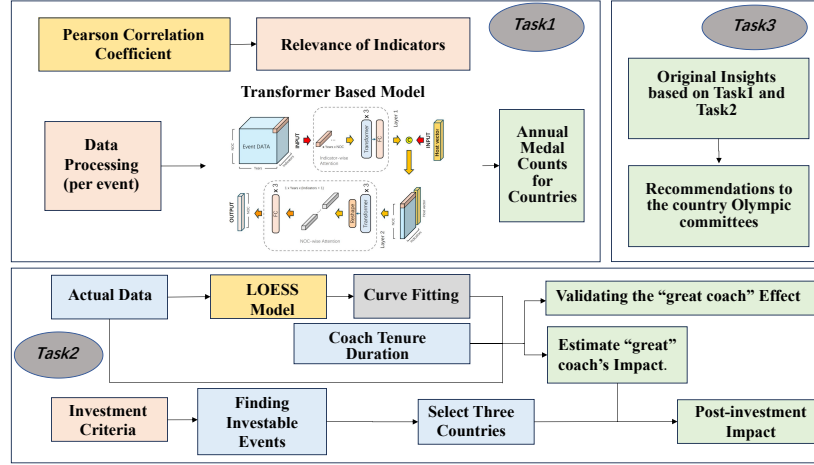


Figure 1: Visualization our work.

2 Assumptions

Considering the complexity of the practical issues and global uncertainties, we propose the following basic assumptions to simplify the problems at hand:

- **Assumption 1:** Assume that the trends in medal counts of countries from past Olympic Games will continue in future Olympics, serving as the basis for predictive models.
- **Assumption 2:** Assume that athletes' future performance correlates with their historical achievements, and the number of participating athletes as well as their event distributions across countries will remain relatively stable.
- **Assumption 3:** Assume that the relationship between Olympic event structures and medal allocations will remain stable in future Games. Changes in event categories or scoring rules are not expected to significantly alter medal distributions.
- **Assumption 4:** Assume that the "great coach" effect has a statistically significant and quantifiable impact on athlete performance and team medal outcomes, with this influence being consistent across countries and sport disciplines.

3 Notations

Some important mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper.

Symbol	Description
E	Set of all events.
S	Set of all sports.
C	Set of all countries.
I	Influence of a coach.
$\hat{v}$	Fitted smoothed medal counts.
MS	Medal scores.

4 Data Preprocessing

We first preprocess the given data, including focusing on only partially trend-compliant data, as well as dealing with data names, duplicates, and missing values.

- ***Limiting Data to Post-World War II Era (1948 Onwards).***

The World War II stands as one of the most significant events in human history, profoundly reshaping the global landscape, including the dynamics of international sports within the Olympic Games [2, 3]. Therefore, to ensure the stability and reliability of our data, this paper only focus on Olympic Games data from 1948 onwards.

- ***Correcting and standardizing data names.***

We discovered some extra spaces after country names (NOCs) in the dataset “summerOly medal counts.csv”, which could lead to incorrect segmentation of countries. We removed these extra spaces and standardized all country names for data accuracy.

Additionally, we identified inconsistencies and errors in event names in the dataset “summerOly athletes.csv”. For example, the event “3×3 Basketball” was incorrectly listed as “3×3 Basketball, Basketball” for athlete Amy Atwell in the 2024 Paris Olympics. Furthermore, the same events, such as “Athletics Women’s 100 metres”, are named differently across several Olympic Games. To ensure consistency, we corrected and standardized these names.

Moreover, for cases where different team names referred to the same country (e.g., “DPR Korea” and “North Korea”), we standardized these names to maintain data consistency.

- ***Removing duplicates and handling missing data.***

We found duplicate entries in the “summerOly athletes.csv” dataset, such as the record of athlete Alfred Munnings participating in Art Competitions appear 25 times in the dataset for the 1948 Olympics. To avoid data redundancy, we removed the duplicate entries. We also discovered missing data in the “summerOly programs.csv” dataset, such as the number of badminton events in the 1988 Olympics. We corrected these value to 0 based on calculation to ensure data integrity.

5 Task 1: Medal Counts Prediction and Analysis

In this section, to fully solve Task 1, we construct a **Transformer based Model** to predict the medal outcomes for **each event**, which fully utilizes the advantage of **attention mechanism** for complex internal relationships and provides a **more fine-grained event-based estimation**. We then select **eight indicators based on Pearson correlation coefficient** as the model input. Then, based on this model, we predict the **medal table in 2028** with prediction intervals and identified the **progressing countries and forecast which countries are likely to win their first medal**.

5.1 Indicators Analysis

5.1.1 Establishment of Indicators

In this part, we identify and summarize the key indicators related to medal counts, and conduct a correlation analysis to validate our findings. Specifically, in order to summarize the factors associated with the total medal counts, and accurately predict the medal standings for the next Olympics, we first visualize the total medal counts of the top ten countries in Olympic Games since 1948. Moreover, we also visualize the relationship among athletes' participation in Olympic events, number of times and their average scores.

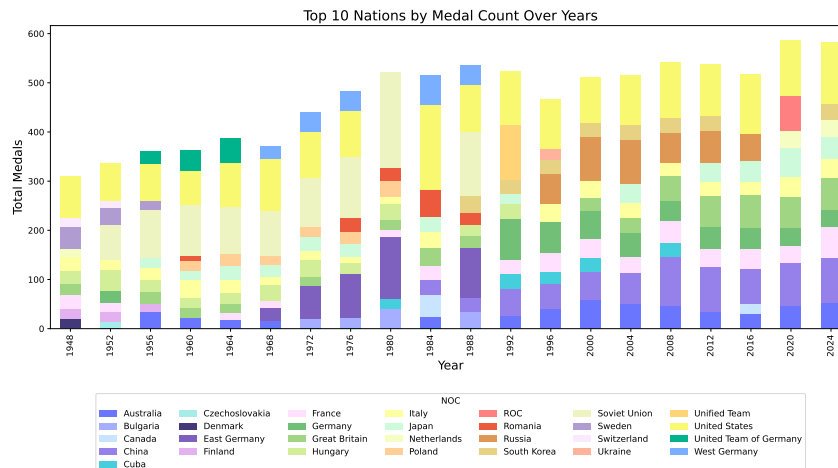


Figure 2: Medal Counts of the top 10 nations.

Figure 2 shows the distribution of medal count for top 10 countries, where each bar represents the totals medal with different colors indicating different countries. The top of each chart highlights trends in medal counts across years. **Specifically, the yellow bars, representing the United States, consistently dominate the rankings over the years, reflecting the country's sustained excellence in the Olympics.** Such success indicates that medal counts are likely influenced by various factors, such as national resources and historical performance. Thus, a nation's medal success appears to be linked to various factors associated with each country.

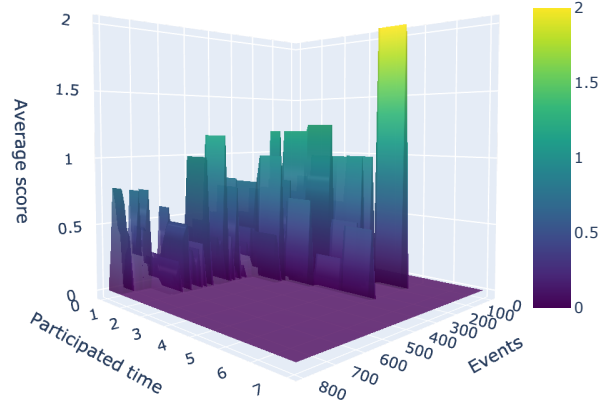


Figure 3: 3D visualization of average score for each athletes with different participated times.

We displayed a 3D visualization of average score for athletes with different participated times in Figure 3, which demonstrates a clear upward trend in the average scores with higher times in the same Olympic event. This suggests that **the competence level of athletes, especially for those star athletes, developed through repeated participation, plays a significant role in determining a country's medal success.**

5.1.2 List of Relevant Indicators

We analyze the factors behind these trends displayed in figures above, and identify six key indicators, processing the data for subsequent correlation analysis. They are as follows:

- x_{truth} : *Medal scoring for each event.*

For the purpose of later modeling analysis, we calculate medal scores as follows:

$$x_{truth} = \begin{cases} 3, & \text{for gold medal,} \\ 2, & \text{for silver medal,} \\ 1, & \text{for bronze medal,} \\ 0, & \text{for no medal.} \end{cases} \quad (1)$$

- x_{sex_type} : *Sex of athletes.*

Some countries may perform better in either men's or women's sports, leading to more medal counts. For instance, North Korea has demonstrated notable success in women's diving, winning a silver medal, while its men's diving team has not achieved similar prominence.

In the correlation analysis, we categorized the input data by gender as follows:

$$x_{sex_type} = \begin{cases} 0, & \text{for males,} \\ 1, & \text{for females,} \\ 2, & \text{for mixed-gender events.} \end{cases} \quad (2)$$

• x_{com_score} : *Athletes' competence level.*

High-level athletes significantly increase the likelihood of medal success. To quantify x_{com_score} , we use a quantified score to measure an athlete's level, which is based on their past Olympic performances and the probability of participating in the next Olympic Games, and it is defined as follows:

$$x_{com_score} = D_score (MS_{past}, T) \times P_{next}(T), \quad (3)$$

where $D_score (MS_{past}, T)$ denotes the composite scores based on an athlete's historical performance, T represents the number of Olympic Games the athlete has participated in, and $P_{next}(T)$ denotes the probability of the athlete participating in the next Olympic Games.

Specifically, the formula for $D_score (MS_{past}, T)$ is expressed as follows:

$$D_score (MS_{past}, T) = \sum_{t=1}^T MS_t \cdot e^{-k \cdot (T-t)}, \quad (4)$$

where k is the decay factor 0.1, MS_{past} stands for the athlete's previous Olympic scores.

To calculate $P_{next}(T)$, we first determine the probability of participation in the T -th Olympic Games, denoted as $P_{(T)}$:

$$P_{(T)} = \frac{N_T}{N_{total}}, \quad (5)$$

where N_T denotes the number of athletes who participated in the T -th Olympic Games, and N_{total} denotes the total number of athletes who competed in that year.

Next, we calculate the probability that an athlete who has participated in the T -th Olympic Games will compete in the next Olympic Games, which is expressed as: $P_{next}(T)$:

$$P_{next}(T) = P(T+1 | T) = \frac{P(T+1)}{P(T)}, \quad (6)$$

Furthermore, we introduce the indicator x_{team_score} based on x_{com_score} , which represents the overall skill level of a team in team-based events. It is calculated by summing the scores x_{com_score} of each athlete in the team.

• x_{total_score} : *Countries' strength in specific events.*

A country's strongest sport typically yields more medals, driven by its long-standing tradition, well-established athlete development programs, and robust government support. To quantify this indicator, we denote it as:

$$x_{total_score} = \frac{MS_T}{6}, \quad (7)$$

where the denominator 6 represents the total score, which includes points for gold, silver, and bronze medals.

• x_{host} : *Host country advantage.*

The host country often benefits from the home advantage, as athletes are more accustomed to local climate, food, and other environmental factors [5], making the "home-field effect" a notable

characteristic in multiple Olympic Games. In the following analysis, we assign “1” to the host country and “0” to all other countries.

- x_{team_num} : *Total number of athletes of a country.*

The more athletes a country sends to the Olympics, the greater its chances of winning medals. Furthermore, a larger team may contribute to improved team dynamics, motivation, and overall support. In the following correlation analysis, we input the number of athletes each country sends to the Olympics each year.

Finally, before conducting the correlation analysis of the indicators, we apply **Min-Max normalization** to scale the data. It ensures that all variables are on the same scale, which helps improve the accuracy and efficiency of the analysis. The normalized value X_{norm} is expressed as:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}, \quad (8)$$

where X represents the original data, X_{min} and X_{max} are the minimum and maximum values, respectively. X_{norm} lies within the range $[0, 1]$.

5.1.3 Correlation Analysis

To verify whether there is a correlation between the six selected indicators and the medal counts, we introduce **Pearson correlation coefficient** [4] to assess the strength and direction of the linear relationship between each two variables. The formula for correlation coefficient r is defined as:

$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y})^2}}, \quad (9)$$

where X_i and Y_i are two indicators we listed, $\bar{X}$ and $\bar{Y}$ represent the average values of X_i and Y_i , respectively.

The results of the correlation analysis in all sports are shown in Figure 4. In the heatmap, darker colors indicate higher correlation coefficients, representing stronger correlations. We observe that all indicators show a relatively strong positive correlation with x_{truth} , among which x_{total_score} , x_{team_num} , and x_{com_score} are particularly significant.

To further validate the correlations of the indicators across different sports, we visualize the correlation coefficients for two different representative sports. They are athletics, which is popular and widespread, and volleyball, which is team-based, as shown in Figure 5.

In athletics, x_{total_score} exhibits the strongest correlation with x_{truth} . Following this, x_{com_score} and x_{team_num} also **show relatively strong positive correlations**. These findings align with the nature of athletics, where a country’s strength in an event and individual athlete performances are often combined to determine overall success. Larger teams or those with stronger capabilities are likely to have a competitive edge. In volleyball, x_{total_score} and x_{com_score} (represented as x_{team_score} for team sports) exhibit a strong positive correlation with x_{truth} . Additionally, the correlation between x_{host} and x_{truth} is significantly stronger compared to other sports, suggesting that the home advantage may have a greater impact on team-based sports like volleyball.

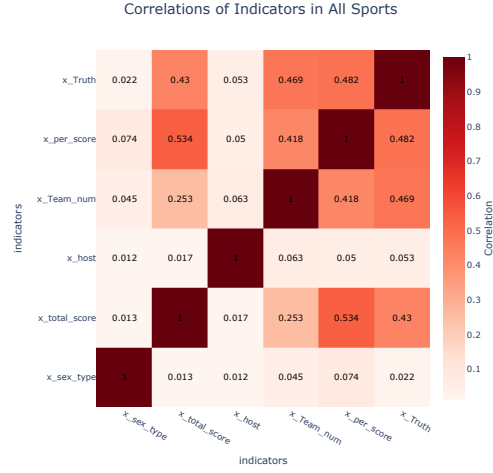
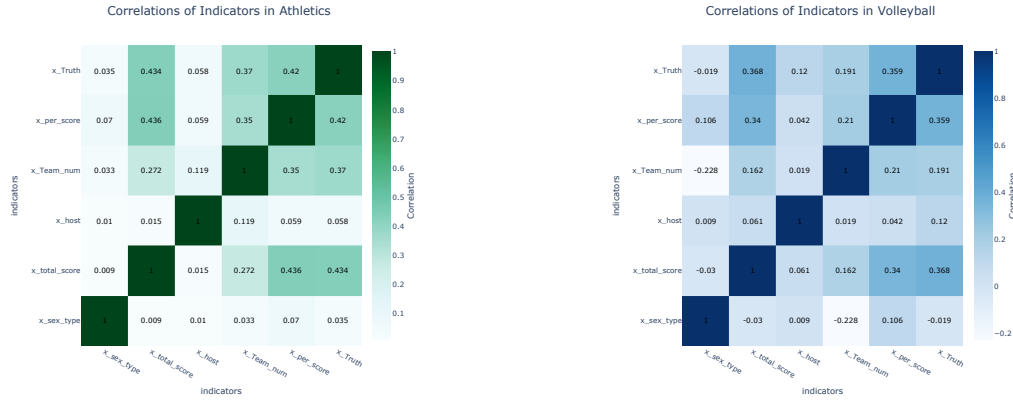


Figure 4: Correlations of indicators in all sports.



(a) Correlations of indicators in athletics.

(b) Correlations of indicators in volleyball.

Figure 5: Correlations of indicators in particular events.

5.2 Multi-dimensional Prediction Based on Transformer Architecture

To predict medal counts for the next Olympic Games, we analyze historical medal data and additional information from the datasets. **The Transformer model** [6], with its self-attention mechanism and parallel processing capabilities, is particularly effective for capturing complex sequential relationships, making it an ideal choice for generating accurate predictions based on historical trends.

5.2.1 Dataset Construction

Due to differences in years and events, the index positions of NOC may vary. If we perform padding operations on the empty positions at this point, the temporal sequence information will be lost. However, our medal prediction model needs to retain the temporal sequence information. Therefore, we first traverse all the data in a single event to count the index positions of all participated

NOC, generating a tensor containing this positional information. We then randomly pad this index vector to achieve certain length of 90, which is the maximum amount of participated countries across all events. Note here, as the padding data is extremely sparse, hence, we use random padding to ensure a equal distribution for each NOC position. Subsequently, based on this padded NOC index tensor, we align the indices of all countries to the same position. After this operation, we carry out the padding. In this way, we can ensure that our data will not lose the temporal. The specific operation is shown in Figure 6.

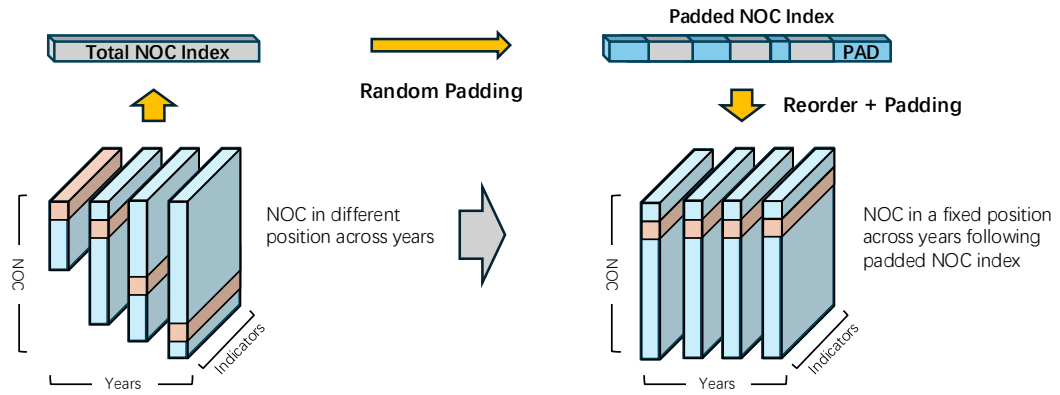


Figure 6: Overveiw of random padding.

5.2.2 Fundamentals of the Transformer Architecture

The Transformer model is a deep learning architecture that processes data in parallel using self-attention. It consists mainly of an Encoder and a Decoder, each of which contains a Self-Attention mechanism and a Feed-Forward Neural Network. The Self-Attention mechanism allows the model to dynamically adjust the attention weights to capture long-range dependencies. The model structure is shown in Figure 7.

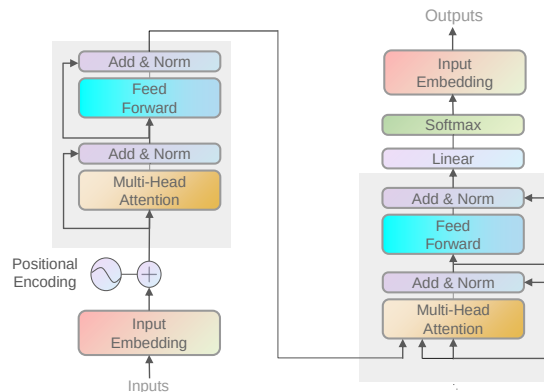


Figure 7: Transformer model structure.

The self-attention mechanism is the core module of the Transformer, designed to capture relationships between different positions in the input sequence. The main steps are as follows:

Step 1: Each input vector is linearly transformed to generate the Query (Q), Key (K), and Value (V) matrices:

$$Q = XW^Q, \quad K = XW^K, \quad V = XW^V, \quad (10)$$

where X is the input vector, W^Q , W^K , and W^V are the linear transformation matrices.

Step 2: The similarity between the query and key is computed using the dot product, scaled to prevent large values, and passed through the softmax function to obtain attention weights:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V, \quad (11)$$

where QK^T represents the dot product between the query and the key, d_k is the dimension of the key vector, used for scaling to avoid overly large values.

Step 3: The attention weights are used to compute a weighted sum of V , generating the attention output.

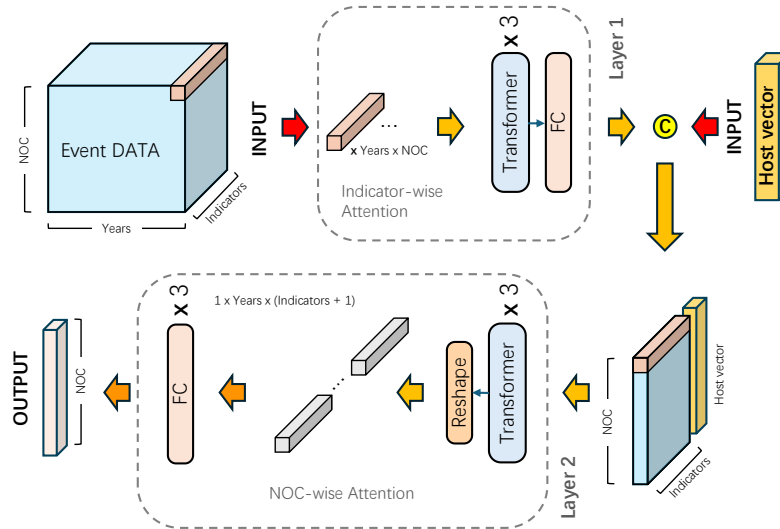


Figure 8: Model one overview.

5.2.3 Establishment of the Network Structure

To decode Olympic medal success and identify the pivotal factors influencing national performance, we divided the data according to each event in the Olympic Games and constructed a multi-dimensional data cube model with three dimensions: years, NOC, and indicators. This model integrates these three-dimensional data to capture the complex dynamics of Olympic medal counts. The specific overview of this model is shown in Figure 8, which consist of two layers of transformer each focusing on distinct dimensions.

Specifically, in Layer 1, we input indicator data for each country and year, employing a three-layer stacked Transformer model to capture intricate relationships among six key indicators. The

output is then compressed into two-dimensional data through a fully connected layer, effectively reducing the year dimension while preserving essential information. To account for the host country effect, we concatenate the compressed data with a host tensor, enhancing the model's ability to capture the increase of medal counts often seen in host countries.

In Layer 2, we use another three-layer stacked Transformer to model competitive relationships among countries, generating a one-dimensional tensor that encapsulates global Olympic dynamics. Finally, a three-layer stacked fully connected layer maps this tensor to a 90×1 tensor (NOC x Score), representing the final score (Gold:3; Silver:2; Bronze:1; None:0) for each country and enabling comprehensive medal predictions.

5.2.4 Training Methods for Network Structures

Prior to model training, we filtered Olympic events to retain only those consistently featured across multiple Games, excluding transient or irregularly held events. Long-standing events have rich historical data for capturing trends and accurate predictions, improving data quality and model robustness.

We train our model using 120 Epoch and split the dataset into training and testing sets with a ratio of 9 to 1. This division ensures that our model learns the underlying patterns during training while being evaluated for generalization performance on the test set.

Additionally, during each training iteration, we compute the error between the model's predictions and the ground truth using a loss function. The model parameters are then updated via gradient descent to minimize this error and improve data fitting.

To ensure that the model's prediction results comply with the actual medal distribution rules, we design two loss functions: Sum Loss L_{sum} and Conditional Loss L_{CMSE} .

L_{sum} aims to ensure that there are at most three medals in an event: one gold, one silver, and one bronze. In team events, a team winning a medal is counted as one. This goal is achieved by traversing the predicted scores of all countries, calculating the total score for each country, and using the L2 norm to constrain the total score to be no more than 6. The specific formula is as follows:

$$L_{\text{sum}} = \sum_{i=1}^n y_i - 6, \quad (12)$$

where y_i is the predicted score for country i .

L_{CMSE} aims to mitigate the impact of sparse matrices. In particular, when the label is 0, the model tends to push the predicted values towards 0, resulting in overly sparse prediction results. By introducing the indicator function $\mathbf{1}$, when the true label is 0, the value of L_{CMSE} is reduced by a hyperparameter ω (set to 0.005 here), thus lessening the influence of labels being 0. The specific formula is as follows:

$$L_{\text{CMSE}} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \mathbf{1}, \quad (13)$$

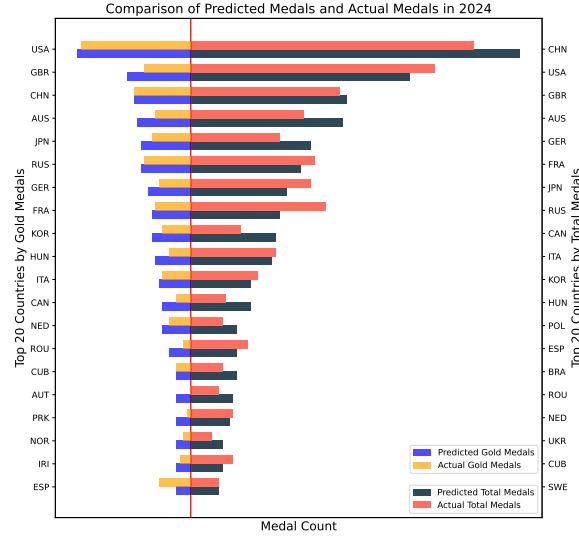


Figure 9: Comparison of predicted medals and actual medals in 2024.

where n is the number of countries, and $\mathbf{1}$ is expressed as:

$$\mathbf{1} = \begin{cases} \omega, & \text{if } \hat{y}_i = 0, \\ 1, & \text{else,} \end{cases} \quad (14)$$

In addition, to balance the effects of the two loss functions, we introduce the hyperparameter α to adjust the weights. The final loss function is defined as:

$$\text{Loss} = L_{\text{CMSE}} + \alpha L_{\text{sum}}, \quad (15)$$

where α is a hyperparameter used to control the weight balance between L_{sum} and L_{CMSE} . By adjusting α , we can flexibly regulate the influence of the two loss functions on model training, thereby optimizing the model's prediction performance. Here, we set it to 0.0004.

5.3 Analysis of Model Results

5.3.1 Estimates of the Precision of our Model

To evaluate the precision of our model, we predicted the medal counts for the top 20 countries in the 2024 Olympics based on data from previous Games. These predictions were then compared with the actual medal outcomes in 2024, as visualized in Figure 12, which displays both the predicted and actual total medal counts and gold medal counts for each country. As shown, **the predicted values closely align with the actual results**, demonstrating our model's reliability.

In addition, we calculated the MSE Loss value. As the model training progressed, the Loss consistently converged, reaching a final value of **0.5577557682991028**, as shown in Figure 10. This

low and converging value attests to the model's effectiveness, indicating its ability to accurately predict by minimizing the gap between predictions and actual values.

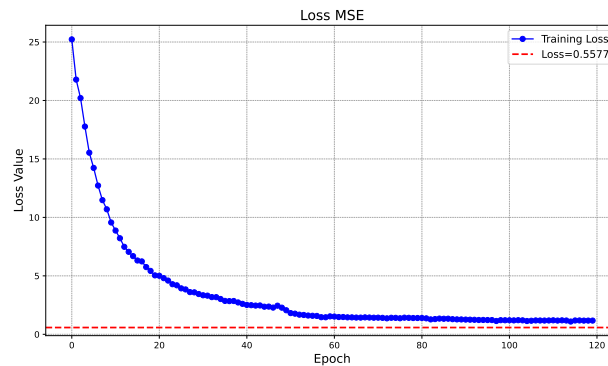


Figure 10: MSE Loss.

5.3.2 2028 Medal Table Predictions

Based on the aforementioned model, we present in Figure 11 the top ten countries ranked by the total number of medals in the model's prediction results, along with the specific medal details for each country. **Our prediction indicates that in 2028, the United States will win the largest total number of medals, reaching 94, followed by China with 84 medals. Moreover, the United States and China will also rank among the top two countries in terms of the number of gold medals won.**

	NOC	Country				Total
1	USA		30	35	29	94
2	CHN		26	28	30	84
3	RUS		17	20	10	47
4	GBR		14	13	6	33
5	FRA		12	13	2	27
6	AUS		11	8	7	26
7	ITA		11	4	4	19
8	JPN		9	3	6	18
9	KOR		8	4	5	17
10	ESP		7	4	4	15

Figure 11: Predicted medal table for 2028.

Our **prediction-interval calculation method** assesses each event by selecting the top-four

scoring countries. Analyzing their score variance helps evaluate prediction confidence: a smaller variance means higher confidence. We use the ideal-condition variance as a benchmark to calculate the error probability for measuring prediction accuracy. Ideally, the top four countries in each event would have ideal scores of 3,2,1,0. The variance was calculated to be 1.25. We define the set of the top four highest scoring countries in each event as T_e . The specific steps are as follows:

Step 1: Calculate the average score of the top four countries μ_e .

$$\mu_e = \frac{1}{4} \cdot \sum_{i=1}^4 MS_{e,i}, \quad (16)$$

Step 2: Perform variance calculations σ_e^2 .

$$\sigma_e^2 = \frac{1}{4} \sum_{i=1}^4 (MS_{e,i} - \mu_e)^2, \quad (17)$$

Step 3: Calculate the probability of error P_c^e for a single event.

$$P_c^e = \begin{cases} 1 - \frac{\sigma_e^2}{1.25}, & \text{if } c \in T_e \text{ and } \sigma_e^2 \leq 1.25, \\ 0, & \text{otherwise} \end{cases}, \quad (18)$$

Step 4: Sum the error probabilities for all events and round down, obtaining the prediction interval F_c :

$$F_c = \left\lfloor \sum_e P_c^e \right\rfloor. \quad (19)$$

If a country's medal counts in 2024 falls below the predicted range for 2028 by our model, it indicates that the country's medal-winning situation is most likely to improve, such as the GBR shown in Figure 11. Conversely, if its medal count in 2024 exceeds the predicted range for 2028, it suggests that the country's medal-winning situation is likely to worsen, like the USA.

Table 2: Countries predicted to win medals.

NOC	Medal Types
ANT	bronze
GBS	bronze
ESA	bronze
CAF	bronze
MAW	bronze

In addition, Table 2 shows the countries that are predicted by our model to win medals for the first time in 2028 and the types of medals they will receive.

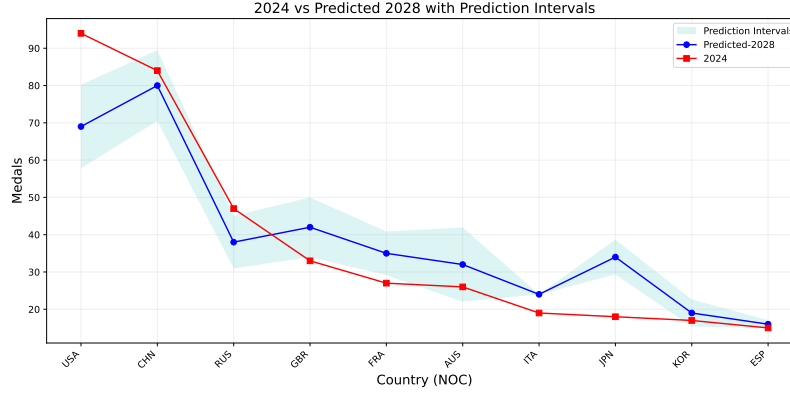


Figure 12: Medal Prediction vs Actual Values.

5.4 Relationship Between Events and Medal Counts

In order to explore the relationship between events and medal counts, we introduce the “event yearly score rate” $P(c, e)$, which is defined as:

$$P(c, e) = \left(\sum_{y \in Y} \frac{MS(c, y, e)}{\sum_{c' \in C} (c', y, e)} \right) / 20, \quad (20)$$

Through calculations, we visualize the average number of points earned per session as a percentage of the total number of points earned by selected countries in a given event, which is shown in Figure 13.

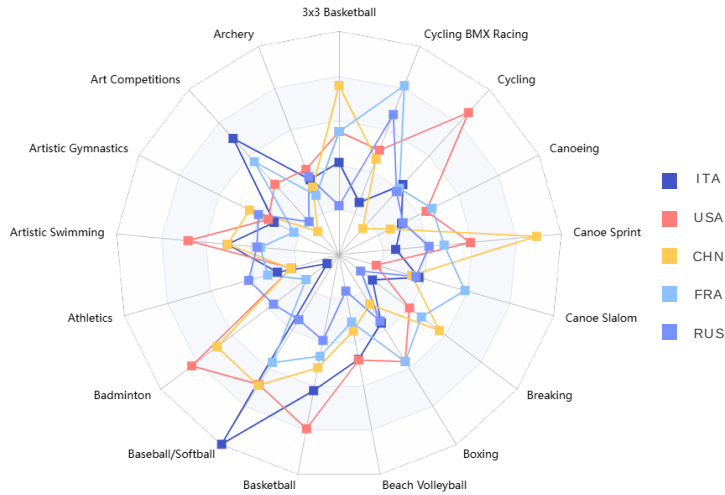


Figure 13: Event yearly score rate of different countries.

It can be observed that different countries have events that they excel at scoring points in. For example, RUS is very dominant in baseball/softball, while the US leads in scoring in basketball.

Given that the host country has the option of adding sports to the Olympic Games, and the option of adding a dominant sport can be based on the proportion of the total number of points

that the country earns on average per sport, increasing the country's total number of points scored at particular Olympic Games. Specifically, the change in a country's medal score in an event $\delta MS(c, e)$ can be expressed as:

$$\delta MS(c, e) = n * P(c, e) * 6, \quad (21)$$

where n indicates the number of additions to the event. Taking China as an example, when China chooses to add 2 Canoe Sprint events, based on the historical scoring ratio of 0.4304, it can be estimated that these n events will add a score of **5.16** for China.

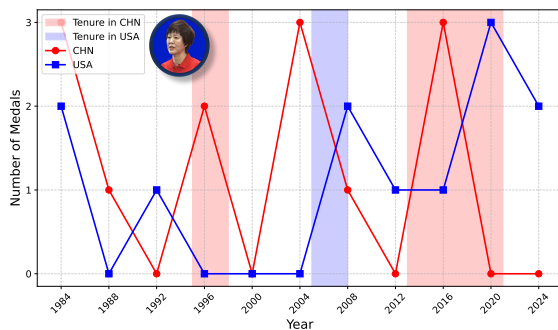
6 Task 2: Great Coach Effect on Medal Counts

To assess the impact of great coaches on medal counts and guide investment in national sports, we first **visualize** Olympic medal achievements under **Lang Ping and Béla Károlyi**, suggesting a **potential “great coach” effect**. Then we apply the Local Weighted Scatterplot Smoothing (LOESS) Model to **confirm this effect** and quantified the coaches' influence by averaging the differences between actual and fitted medal scores. Finally, we establish **three criteria to evaluate “revival potential”**, identifying countries and sports for investment in great coaches.

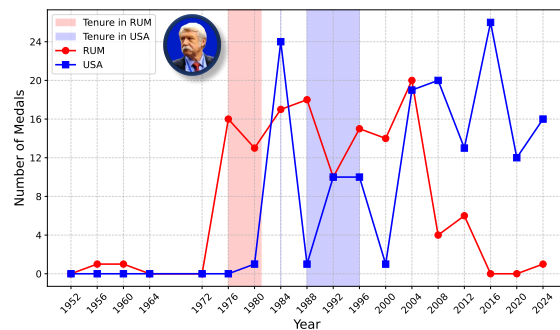
6.1 Medals Changing Trends

6.1.1 Possibility of Great Coach Effect

To analyze **whether** the presence of a “great coach” impacts the medal counts, we visualize the Olympic medal scores in women's volleyball for China and the United States, and in gymnastics for Romania and the United States, investigating the potential existence of the “great coach” effect. Figure 14 visualizes the results, **with the x-axis representing Olympic years and the y-axis showing medal scores based on point allocation**. The blue and red lines depict the medal scores of the two countries, respectively, while the purple and red shaded areas indicate the coaches' tenure periods in each country.



(a) Women's Volleyball Medals: China vs. USA.



(b) Gymnastics Medals: Romanian vs. USA.

Figure 14: Visualization of medal scores under different coaches' tenure.

Specifically, in Figure 14(a), results show that during **Lang Ping's tenure**, both countries reached better or peak performance in medal counts. Especially when coaching the **United States team**, the average score rose significantly from 1 point to 2 points after 2008, showcasing a consistent overall improvement for **the USA women's volleyball team** brought by Lang Ping. Similarly, when coaching **the Chinese team**, the average score increased from 1 point to 1.6 points, winning a silver medal in 1996 and a gold medal in 2016. These results show a **significant positive effect** of Lang Ping in improving team performance.

Similarly, Figure 14(b) demonstrates that during **Béla Károlyi's coaching tenure**, both the United States and Romania achieved remarkable success in gymnastics. For example, **the United States** won consecutive Olympic medals in 1992 and 1996, while Romania secured medals in 1976 and 1980. Moreover, both countries exhibit a **strong tendency** to be influenced by the coach. This is particularly evident for the United States, which experienced a notable improvement during Béla Károlyi's coaching period, followed by a **significant decline** in 2000 after his tenure ended. These results suggest the significant impact of a "Great Coach" effect, which **significantly influences** a country's Olympic performance as the coaching tenure begins and ends.

6.1.2 Trend Decomposition Model Based on LOESS

To further **validate** the actual presence of this effect, we develop a model to assess its impact on a country's medal counts during periods when renowned coaches are in charge. By **comparing these results with those from periods without the coaches**, we can gain a clearer understanding of how the "great coach" effect influences the **success of the sports**.

• Locally Weighted Scatterplot Smoothing (LOESS).

To better assess the impact of coaching on national sports medal counts, it is essential to analyze trends in a country's medal performance. However the raw medal data **exhibits a discrete distribution** with significant fluctuations, which presents challenge to access a clear trend. Hence we employ the **LOESS** [7] model to estimate the trend by fitting the data distribution. LOESS is a non-parametric regression method that smooths the data through locally weighted regression, effectively capturing the local trends of the nonlinear relationships within the data. The main formula of LOESS is as follows:

Step 1: For the time of the Olympics t_i , a weight based on the distance is assigned to the data points in its neighborhood, and it is given by:

$$W_i = \text{diag} \left(K \left(\frac{|t_i - t_j|}{h} \right) \right), \quad (22)$$

where K is the weight function, h is the bandwidth, t_i is the current year point, and t_j is the neighboring years based on their distance from the current year.

Step 2: For each year t_i , we perform a weighted regression on the neighboring years, minimizing the weighted sum of squared errors. The specific formula is as follows:

$$\hat{v}_i = \arg \min_{\beta} \sum_j (v_j - T_j \beta)^2 W_{ij}, \quad (23)$$

where T_j is the feature vector of the j -th data point, β is the regression coefficient, representing the fitting parameter, v_j is the target value of the j -th data point, and W_{ij} is an element in the weight matrix, representing the influence of data point t_j on t_i .

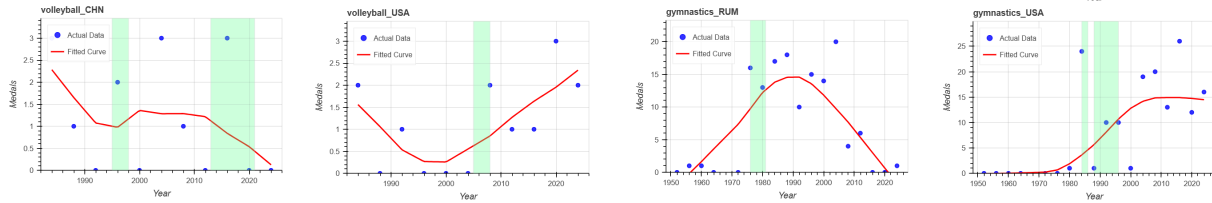
Step 3: Once the regression coefficients are calculated, the predicted medal counts $\hat{v}_i$ for t_i is:

$$\hat{v}_i = T_i \hat{\beta}_i, \quad (24)$$

where $\hat{\beta}_i$ is the regression coefficient obtained through weighted regression.

• Visualization and Analysis of Results.

We visualize the fitted Olympic medal trends for the women's volleyball teams of the United States and China, as well as the gymnastics teams of the United States and Romania. The results are shown in Figure 15, where the fitted curves are represented by red lines, and actual medal scores for each events are displayed with blue dots. The coaching tenures of Lang Ping and Béla Károlyi in their respective countries are indicated by green-shaded areas.



(a) Chinese women's volleyball team. (b) USA women's volleyball team. (c) Romanian gymnastics team. (d) USA gymnastics team.

Figure 15: Visualization of the fitted and actual results.

In this figure, we observe that the the fitted red curves align closely with most blue dots, showcasing a accurate medal trend. Notably, within the green-shaded areas, both the blue dots and the red curves exhibit noticeable deviations, suggesting a correlation between the presence of these renowned coaches and increased medal counts.

Specifically, in Figure 15(a), China's women's volleyball team won silver and gold medals during Lang Ping's tenure, and Figure 15(b) shows a long-lasting positive effect for the US team on medal winning across multiple Olympics after her coaching. Similarly, Figure 15(c) reveals that Béla Károlyi's coaching of Romania enhanced its gymnastics, with good performances in consecutive Olympics, but the effect declined after three Games. In contrast, in Figure 15(d), his coaching of the US improved its gymnastics medal results with a lasting impact. Moreover, the coach effect varies by country, possibly due to culture and sports investment. These findings support the significant "great coach" effect on boosting a country's medal-winning odds, hence, suggesting the necessity of coach investment.

• Quantifying the Impact of Coaching.

To quantify the contribution of the "great coach" effect to medal counts, we establish a impact score based on how much a coach effects the trend. Specifically, we first calculate the difference between the actual medal results and the fitted trend. Then we **sum these differences** during the

coaches' tenures in different countries, and finally average the results to determine the influence attributed to the "great coach." The influence of a coach I is defined as follows:

$$I = \frac{\sum_{i=1}^N \sum_{j=1}^{T_i} (MS_{i,j} - \hat{v}_{i,j})}{\sum_{i=1}^N T_i}, \quad (25)$$

where N represents the number of countries in which the coach has coached, T_i denotes the number of coaching terms in the i -th country, $MS_{i,j}$ represents the medal scores won by the coach in the j -th event within his or her tenure of the i -th country, and $\hat{v}_{i,j}$ is the the predicted value of the fitted data that we derived above. Based on this formula, we calculate each I for Lang Ping and Béla Károlyi as summarized in Table 3, which shows a strong influence of both coach that effects the fitted medal curve.

Table 3: Quantified coaching impact for Lang Ping and Béla Károlyi.

Coach	$I \uparrow$
Lang Ping	0.949142509
Béla Károlyi	3.994157541

6.2 Evaluation of Future Prospects

6.2.1 Analytical Model for Revival Potential

• Investment Criteria in Great Coach Sports.

Based on the above analysis, we have confirmed the **existence** of the "great coach" effect and quantified its impact. **Great coaches can significantly enhance a country's performance in specific Olympic sports, enabling breakthroughs even in historically underperforming disciplines.** Therefore, strategically applying the "great coach" effect is critical for countries seeking to **improve their Olympic outcomes**, particularly those that have yet to **achieve satisfactory results**.

We select countries based on their "**revivable potential**" in specific sports, defined as disciplines where a country historically achieved **strong results** but has since experienced a significant decline in performance. For such cases, we provide strategic recommendations for **investing in great coaches** to reignite competitiveness.

Specifically, we define three criteria for assessing "revivable potential":

Sport-Level Dominance (DOM(c, s)).

A country's medal scores in a specific sport must **exceed 50%** of the global total in a single Olympic Games. The formula for its calculation is:

$$\text{DOM}(c, s) = \frac{MS(c, s, y)}{\sum_{c' \in C} MS(c', s, y)} \geq 0.5, \exists y \in Y, \quad (26)$$

where c denotes a country, s denotes a sport, MS is the country's medal scores, and y represents the time of a particular Olympic session.

Event-Level Consistency ($\text{Streak}(c, e)$).

The country must win Olympic gold medals in a specific event category for **two consecutive Games**. The formula is as follows:

$$\text{Streak}(c, e) = \exists e \in E, \exists y \in Y \setminus \{y_{\text{end}}\}, G_{c,e,y} \geq 1 \wedge G_{c,e,y+1} \geq 1, \quad (27)$$

where e represents a specific event, E is the total events, y_{end} denotes the most recent year a country participated in the event, and $G_{c,e,y}$ indicates the number of gold medals won by the country in event e during year y .

Declining Performance Trend ($\text{Decline}(c,k)$).

It means a **measurable downward** trajectory in the country's performance within the sport or event. We first define $\hat{\theta}_1$ as the time coefficient estimate, and conduct a t-test [8] for it as follows:

- **Null Hypothesis** $H_0 : \theta_1 \geq 0$: No significant downward trend.
- **Alternative Hypothesis** $H_1 : \theta_1 < 0$: Significant downward trend exists.
- p -value: Controls the risk of Type I error (false positive) to ensure the reliability of conclusions.

Then we calculate $\text{Decline}(c,k)$. k is the object of the calculation, which can be either a sport or an event.

$$\text{Decline}(c, k) = (\hat{\theta}_1^k < 0) \wedge (p\text{-value}(\hat{\theta}_1^k) \leq \alpha), \quad (28)$$

where α is the significance level.

Additionally, countries requiring investment in great coaches must meet **at least** one of the first two criteria while mandatorily **satisfying the third criterion**, which is expressed as:

$$\text{Target}(c, k) = (\text{DOM}(c, s) \wedge (\text{Decline}(c, s))) \vee (\text{Streak}(c, e) \wedge \text{Decline}(c, e)). \quad (29)$$

6.2.2 Suggestions for Three Countries' Sports

Based on the analysis above, we use Formula 29 to identify countries **suitable for investing** in "great" coaches. By integrating the fitted data from the LOESS model and the quantified impact values of the "great coach" effect, we derive specific medal scores for particular sports or events in these countries with "great" coaches and visualize the results.

We select the following sport or event for three countries:

For Brazil, we recommend investing in a great coach for its **women's volleyball event**. Similarly, for Cuba, we suggest the recruitment of a top-tier coach for its **women's volleyball team**. For China, we propose hiring an exceptional coach for its **gymnastics program**. The following figures illustrate

Table 4: Recommendations for investing “great” coaches in countries.

Country	Sport or Event
Brazil	women’s volleyball
Cuba	women’s volleyball
China	gymnastics

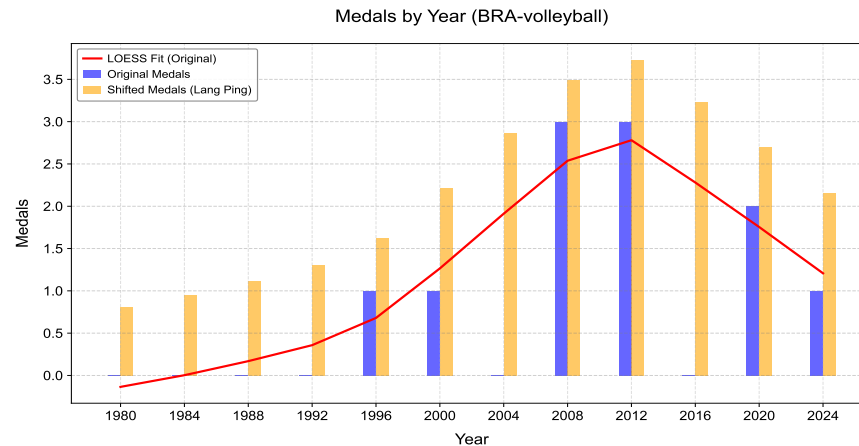


Figure 16: Medal scores of Brazilian women’s volleyball team.

the comparison of medal scores in these respective sports before and after the investment in great coaching, alongside the fitted curves derived from our analysis.

Figure 16 illustrates Brazil’s medal performance in women’s volleyball over the years. It is evident that during Olympic years when the Brazilian women’s volleyball team was guided by great coaches, their performance **significantly improved** compared to periods without such coaching. Additionally, the medal trends under great coaching align closely with the fitted curve, demonstrating the positive impact of strategic coaching investments.

Figure 17 depicts Cuba’s medal performance in women’s volleyball across Olympic years. It demonstrates that during periods when the Cuban team was guided by great coaches, their performance markedly surpassed outcomes achieved without such coaching, particularly evident **in the 2004 Olympics**. Furthermore, the medal trends under great coaching align closely with the fitted curve derive from our analysis.

Figure 18 illustrates China’s gymnastics medal performance over Olympic years. It reveals that the Chinese gymnastics team consistently **outperformed in most cases** under the guidance of great coaches compared to periods without such leadership. Notably, the exceptional performance in 2008 may be attributed to the home advantage during the Beijing Olympics [9].

7 Task 3: Original Insights on Medal Counts

After analyzing the models developed and the results **obtained in Task 1 and Task 2**, we have gained the following **original insights**:

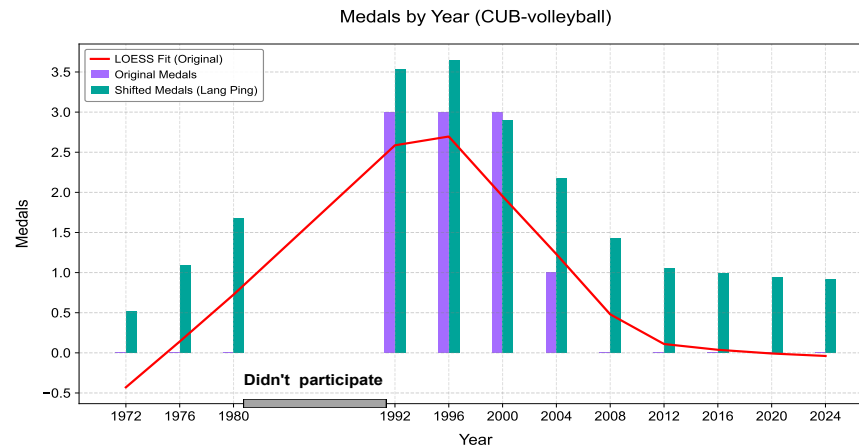


Figure 17: Medal scores of Cuba's volleyball team.

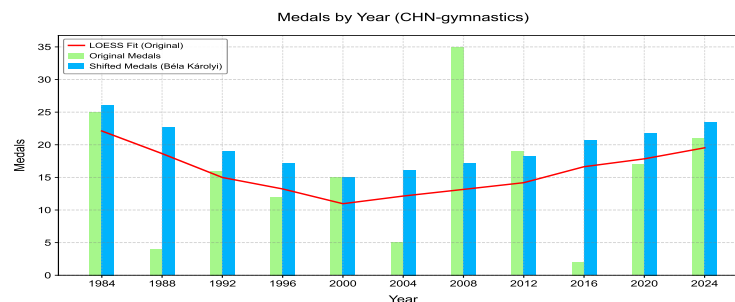


Figure 18: Medal scores of China's gymnastics team.

• Insights from Task 1

Our medal prediction model in Task 1, which is based on the **six core indicators** we proposed, has achieved **excellent prediction results**. This fact inspires us to **pay more attention** to indicators such as **athletes' individual capabilities** and **the host-country advantage**. We recommend the following to national Olympic committees:

- (i) Analyze historical Olympic data using our **indicator-building method** to identify areas for **improvement**.
- (ii) Use our prediction model to **forecast medal changes** in future Olympic Games and **adjust resource allocation** and **training strategies** for corresponding events and athletes accordingly.

• Insights from Task 2

In Task 2, we have revealed the "Great Coach Effect" and developed a model to **quantify coaches' influence** and **criteria for identifying promising sports events**. Combining these **findings**, we offer the following suggestions to national Olympic committees:

- (i) Utilize our criteria to discover promising sports events in the country and **focus on their development**.

- (ii) Analyze the coaching data of domestic and international coaches using our model to quantify their influence and **seek out outstanding coaches**.
- (iii) Use the model to **simulate** the changes in the number of medals for a promising sports event after hiring a corresponding coach, so as to **determine the necessity of investment**.

Given the excellent performance of our models, we believe that by implementing the above insights and corresponding recommendations to national Olympic committees, **the advantages can be extended to both athletes and sports events**, leading to a significant improvement in a country's Olympic performance.

Particularly for countries with **limited resources or without obvious advantageous events**, using our models can provide a **correct development direction** for their Olympic endeavors and enable them to perform better in future Olympic Games.

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